

EDUCATION

School of Computer Science, Wuhan University

Wuhan, China

*PhD's degree in Computer Science and Technology*2021.09 - 2026.06 (*expected*)

- Advisor: Prof. Weiwei Liu
- Research area: Statistics Learning Theory, Out-of-distribution Learning, Representation Learning, Trustworthy Machine Learning
- Average Score: 91.1/100

School of Computer Science, Wuhan University

Wuhan, China

Bachelor's degree in Software Engineering

2017.09 - 2021.06

- GPA: 3.80/4.00

RESEARCH
INTERESTS**Statistical Learning Theory**

- *PAC Learning*: A statistical framework that bounds the sample size needed to learn a hypothesis that, with high probability, has small error on unseen data.
- *Generalization Theory*: Studies why/when models trained on finite data perform well on new data, providing error bounds via capacity measures (e.g., VC/Rademacher) and algorithmic stability.

Out-of-distribution Learning

- *Out-of-Distribution Generalization*: Methods and theory to train predictors that stay accurate under distribution shift from source domains to unseen target domains.
- *Out-of-Distribution Detection*: Identifies inputs outside the training distribution and enables abstention/calibration or alternative handling.

Representation Learning

- *Contrastive Learning*: Representation learning that pulls semantically similar pairs together and pushes dissimilar pairs apart, producing invariant, transferable embeddings for downstream tasks.

Trustworthy Machine Learning

- *Adversarial Robustness*: Adversarial robustness studies how to make models keep correct predictions when inputs are deliberately and imperceptibly perturbed. It designs training and evaluation methods to reduce performance drops under such worst-case perturbations.

- PUBLICATIONS
1. **Xin Zou**, Xiuwen Gong, Weiwei Liu. **Generalization Bounds for Out-of-distribution Generalization**. In *International Conference on Machine Learning (ICML)*, 2026.
 2. **Xin Zou***, Zhengyu Zhou*, Jingyuan Xu, Weiwei Liu. **A Boosting-Type Convergence Result for AdaBoost.MH with Factorized Multi-Class Classifiers**. In *Neural Information Processing Systems (NeurIPS)*, 2024. Solves a COLT 2014 open problem.
 3. **Xin Zou**, Weiwei Liu. **Coverage-Guaranteed Prediction Sets for Out-of-Distribution Data**. In *Association for the Advancement of Artificial Intelligence (AAAI)*, 2024.
 4. **Xin Zou**, Weiwei Liu. **On the Adversarial Robustness of Out-of-distribution Generalization Models**. In *Neural Information Processing Systems (NeurIPS)*, 2023.
 5. **Xin Zou**, Weiwei Liu. **Generalization Bounds for Adversarial Contrastive Learning**. In *Journal of Machine Learning Research (JMLR)*, 2023.
 6. Jingyuan Xu, **Xin Zou**, Xiuwen Gong, Weiwei Liu. **On Learnability and Disambiguation of Multiclass Partial Concept Classes**. In *International Conference on Machine Learning (ICML)*, 2026.
 7. Youjun Wang, Boqi Li, **Xin Zou**, Weiwei Liu. **Neural Collapse in Multi-Task Learning**. In *International Conference on Learning Representations (ICLR)*, 2026.
 8. Xinsong Ma, Jie Wu, **Xin Zou**, Weiwei Liu. **A Unified Decision Rule for Generalized Out-of-Distribution Detection**. In *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025.
 9. Xinsong Ma, **Xin Zou**, Weiwei Liu. **An Online Statistical Framework for Out-of-Distribution Detection**. In *International Conference on Machine Learning (ICML)*, 2025.
 10. Xiyuan Li, **Xin Zou**, Weiwei Liu. **Residual Network with Self-Adaptive Time Step Size**. In *Pattern Recognition (PR)*, 2024.
 11. Xinsong Ma, **Xin Zou**, Weiwei Liu. **A Provable Decision Rule for Out-of-Distribution Detection**. In *International Conference on Machine Learning (ICML)*, 2024.
 12. Xiyuan Li, **Xin Zou**, Weiwei Liu. **Defending Against Adversarial Attacks via Neural Dynamic System**. In *Neural Information Processing Systems (NeurIPS)*, 2022.

ACADEMIC
SERVICES

Conference Reviewer

- Neural Information Processing Systems (NeurIPS, 2023-2025)
- International Conference on Machine Learning (ICML, 2023-2026)
- International Conference on Learning Representations (ICLR, 2024-2026)
- Artificial Intelligence and Statistics (AISTATS, 2025-2026)
- Association for the Advancement of Artificial Intelligence (AAAI, 2025-2026)
- International Joint Conferences on Artificial Intelligence (IJCAI, 2024)

Journal Reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- Transactions on Machine Learning Research (TMLR)
- Neurocomputing

RESEARCH
GRANTS

The National Natural Science Foundation of China (NSFC)

Fundamental Research Project for Young Professional 2025.01 - 2026.12

The Youth Talents Support Project

Doctoral Student Special Program 2025.01 - 2026.06

AWARDS
AND
HONORS

- **Tianyuan Dic Scholarship**, TYDIC 2025
- **Lei Jun Breakthrough Scholarship**, Wuhan University 2025
- **National Scholarship** (for Ph.D. students), Ministry of Education (China) 2024
- **“Huawei” Scholarship**, Huawei Technologies Co., Ltd. 2023
- **Outstanding Student Scholarship**, Wuhan University 2018-2020, 2023-2025
- **National Scholarship** (for undergraduate students), Ministry of Education (China) 2018